

## PyBaMM as the synthetic-corpus generator, DCIR+RPT+50-cycle input

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Second-life reuse of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost. Per-cell PyBaMM [1] calibration of degradation parameters against measured state-of-health (SoH) trajectories is reliable but requires  $\gtrsim 400$  cycles per cell [2]; below that, the fitted rates are under-determined. Prior second-life work [3] and universal-differential-equation degradation [4] improve calibration *quality* without shrinking the cycling budget. We ask: *can a physics-conditioned neural operator, pretrained on PyBaMM-generated synthetic trajectories, predict the full SoH curve of a new cell from a single characterisation snapshot plus a short cycling window?*

We construct a two-stage pipeline. *Stage 1 — Physics corpus*: starting from PyBaMM’s Prada2013 LFP chemistry with per-cell BOL parameters identified from OCV, GITT, HPPC and RPT on a supplier-B fresh-cell cohort (OCV RMSE 6.8 mV median, GITT  $R^2$  0.9999), we run a Sobol sweep of 300 simulations across five degradation channels —  $k_{\text{SEI}}$ , SEI partial molar volume, lithium-plating exchange current, positive- and negative-electrode LAM rates — producing 300 synthetic SoH trajectories with paired ground-truth parameter vectors  $\theta$ . *Stage 2 — Neural operator*: a DeepONet [5] variant whose branch consumes a five-stream fingerprint (DCIR profile, RPT capacity+IC-peak features, first- $K$  measured SoH values, the PyBaMM parameter vector  $\theta$ , and the cycling protocol) and whose trunk consumes the target cycle number; the output is  $\text{SoH}(n) = \text{SoH}_{\text{init}} - \text{softplus}(\langle b, t \rangle)$ , hard-monotonic by construction. The  $\theta$ -conditioning lets the operator absorb electrochemistry *directly* from the Phase-1 param-ID step rather than reconstructing it from characterisation.

Trained on the 300-sim corpus with  $K = 50$  input cycles, the operator predicts held-out synthetic trajectories to **1.0 pp SoH RMSE median** on a 15% validation split (322k parameters, 500 epochs,  $\sim 10$  s per epoch on a single GPU). The talk will present three extensions being finalised: (i) an extended 300-sim  $\times$  1500-cycle corpus so the operator sees full end-of-life shape, not just early-window fade; (ii) fine-tuning on real cells from all three suppliers to correct any sim-to-real residual; (iii) evaluation of whether the operator’s predictions transfer across three commercial LFP suppliers (anonymised A/B/C) without per-supplier retraining.

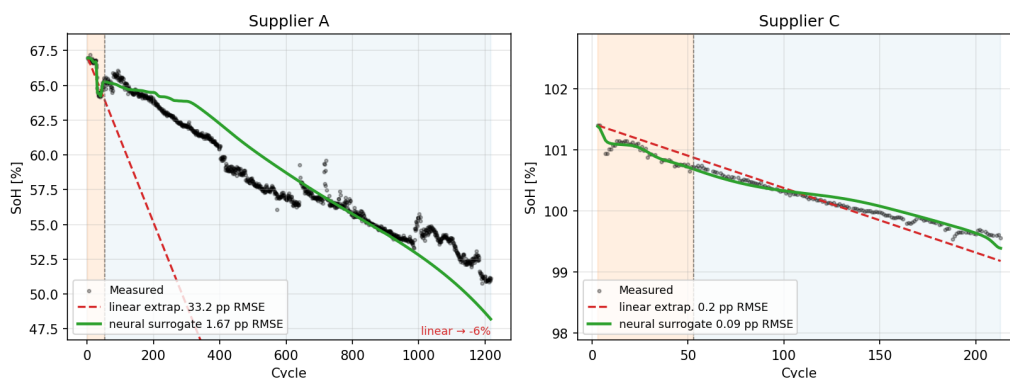


Figure 1: Best-tracking held-out cell per supplier from our earlier PINN baseline ( $K=50$ , ranked by peak deviation across the full held-out window). The neural-operator reformulation retains this input budget while extending output to the full remaining life; results on cells from all three suppliers will be presented at the conference.

**References.** [1] V. Sulzer et al., *PyBaMM*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO<sub>4</sub>-graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025, p. 12. [5] L. Lu et al., *DeepONet: learning nonlinear operators for identifying differential equations*, Nat. Mach. Intell. 3, 218–229 (2021).